2026-2027 – Econ 0107 – Macroeconomics I

Chapter 4 : Dynamic Optimization

(Chapter 4 in Azzimonti, Krusell, McKay, Mukoyama and others's textbook)

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Version 1.0
12/10/2026

Outline

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1. Introduction

Motivation

Solow **model**: saving rate s is exogenous. **This chapter**: endogenize saving via optimization.

- Cass **(1965) and** Koopmans **(1963)**: Added microeconomic foundations to the Solow model $\Rightarrow$ the **optimizing neoclassical growth model (NGM)**.
  - $\times$ Same long-run implications as Solow, but explicit utility $\Rightarrow$ welfare analysis
  - $\times$ Forward-looking decisions $\Rightarrow$ richer policy implications
  - $\times$ With stochastic shocks: foundation of real business cycle theory (Chapter 14)
- Friedman **(1957)**: Individuals are forward-looking – decisions depend on expected future income, not just current income.
- Muth **(1961) and** Lucas **(1972)**: Expectations can be modelled as rational. Forecasts about uncertain future variables, those forecasts are consistent with the true underlying economic model, not arbitrary or systematically wrong.

Two solution methods:

1. **Sequential methods**: maximize over sequences
2. **Recursive methods** (dynamic programming): solve functional equations

2. The Generic Problem

The Dynamic Optimization Problem

$$\max_{\{y_t, x_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t \widehat{\mathcal{F}}(y_t) \quad \text{s.t.} \quad x_{t+1} = h(x_t, y_t), \quad x_{t+1} \in \Gamma(x_t)$$

- y_t: **control variables** (consumption, hours, investment)
- x_t: **state variables** (capital, debt, housing stock)
- $\widehat{\mathcal{F}}$: per-period objective (utility or profit)
- β: discount factor; $1/\beta - 1$: discount rate [Stationarity: between t and $t+h$, discount depends only of h (β^h), not on t]
- h: law of motion linking states and controls
- $\Gamma(x_t)$: feasibility set
- x_0: given (exogenous initial condition)

Existence: Weierstrass theorem requires $\widehat{\mathcal{F}}$ continuous, Γ nonempty, closed, bounded.

Theorem 1 (Weierstrass Theorem)

If $f : S \to \mathbb{R}$ is continuous and $S \subset \mathbb{R}^n$ is nonempty and compact (closed and bounded), then f attains both its maximum and its minimum on S. More precisely, there exist $x^\star, x_\star \in S$ such that

$$f(x_\star) \;\leq\; f(x) \;\leq\; f(x^\star) \quad \text{for all } x \in S.$$

- Note that the supremum $\sup_{x \in S} f(x)$ and the infimum $\inf_{x \in S} f(x)$ always exist as extended real numbers, even without any of these conditions.
- What can fail is that these bounds are *attained*: there may be no point in S where f actually equals its supremum or infimum. The three conditions—continuity of f, closedness of S, and boundedness of S—are each individually necessary.

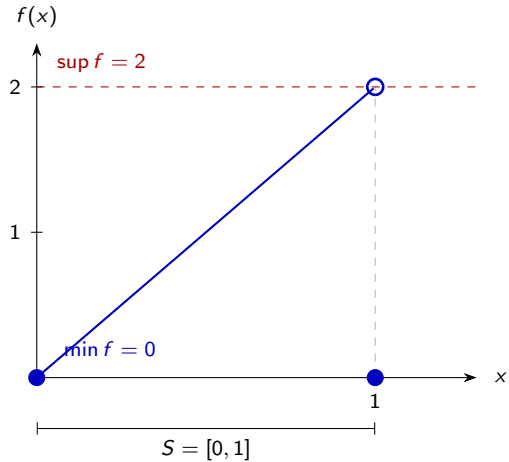
Examples in which Weierstrass Theorem fails

Discontinuous function on a compact set

Let $S = [0, 1]$ and define

$$f(x) = \begin{cases} 2x & \text{if } x \in [0, 1), \\ 0 & \text{if } x = 1. \end{cases}$$

- $\sup f = 2$ not attained (no maximum),
- $\min f = 0$ attained at $x = 0, 1$.



Examples in which Weierstrass Theorem fails

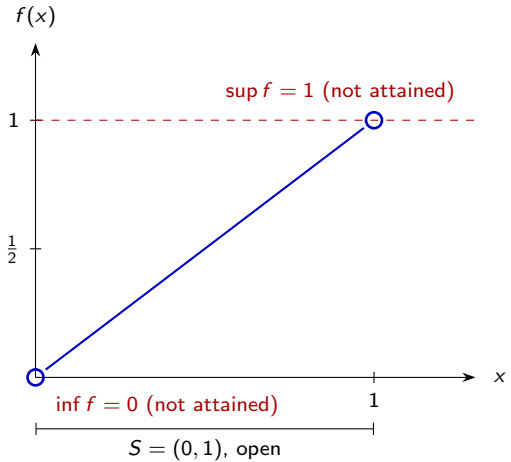
Continuous function on an open set

Let $S = (0, 1)$ (open interval) and define $f(x) = x$.
f is continuous and the domain S is bounded. However, S is *not closed*: the boundary points 0 and 1 are excluded.

$$\sup_{x \in (0,1)} f(x) = 1 \quad \text{(not attained)},$$

$$\inf_{x \in (0,1)} f(x) = 0 \quad \text{(not attained)}.$$

Neither a maximum nor a minimum exists on $(0, 1)$.



Examples in which Weierstrass Theorem fails

Continuous function on an unbounded set

Let $S = [0, \infty)$ and define $f(x) = \dfrac{x}{1+x}$.

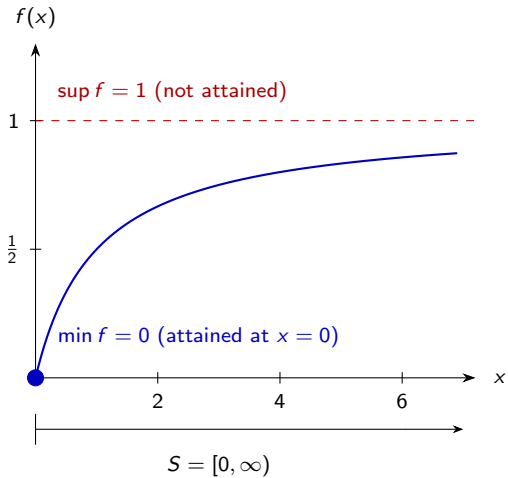
f is continuous, and S is closed and S is *not bounded*.

Minimum is attained: $\min f = 0$.

Supremum satisfies

$$\sup_{x \geq 0} f(x) = \lim_{x \to \infty} \frac{x}{1+x} = 1,$$

but the equation $f(x) = 1$ has no solution: $x/(1+x) = 1$ requires $x = \infty$. Hence f has no maximum on $[0, \infty)$.



Application 1: The Consumption-Saving Model

Agent lives T periods, maximizes $\sum_{t=0}^{T} \beta^t u(c_t)$.
Assumptions on u: strictly increasing, strictly concave, $\lim_{c \to 0} u'(c) = \infty$ (will make $c > 0$, $u'(c) > 0$ (will make the budget constraint always binding).
Budget constraint: $c_t = w_t + (1 + r_t)a_t - a_{t+1}$
Borrowing limit: $a_{t+1} \geq \underline{a}, \quad a_{T+1} \geq 0$

Mapping to generic form:

- Control: c_t; State: a_t
- $h(a_t, c_t) = w_t + (1 + r_t)a_t - c_t$
- $\Gamma(a_t) = [\underline{a},\ w_t + (1 + r_t)a_t]$
- $\widehat{\mathcal{F}}(\cdot) = u(\cdot)$

Application 2: The Neoclassical Growth Model

Same utility. Production: $y_t = f(k_t)$. Capital accumulation:

$$k_{t+1} = (1-\delta)k_t + i_t$$

Resource constraint (closed economy, $i_t = y_t - c_t$):

$$c_t + k_{t+1} \leq f(k_t) + (1-\delta)k_t$$

Planner's problem:

$$\max_{\{c_t, k_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(c_t) \quad \text{s.t.} \quad c_t + k_{t+1} \leq f(k_t) + (1-\delta)k_t, \quad c_t, k_{t+1} \geq 0$$

Mapping: Control: c_t; State: k_t; $\Gamma(k_t) = [0, f(k_t) + (1-\delta)k_t]$, $\widehat{\mathcal{F}}(\cdot) = u(\cdot)$.

Chapter 6: Under perfect competition, the competitive equilibrium solves this planning problem (welfare theorems).

3. Sequential Methods: Finite Horizon

Two-Period Consumption-Saving Model

$T = 2$, $a_0 = 0$, $\underline{a} = 0$, constant r. Budget constraints combine into the **lifetime budget constraint**:

$$c_0 + \frac{c_1}{1+r} = w_0 + \frac{w_1}{1+r}$$

Lagrangian is

$$\mathcal{L} = u(c_0) + \beta u(c_1) + \mu\left\{ w_0 + \frac{w_1}{1+r} - c_0 - \frac{c_1}{1+r} \right\} + \lambda\{w_0 - c_0\},$$

The first-order conditions with respect to c_0 and c_1 are

$$\frac{\partial \mathcal{L}}{\partial c_0}: \quad u'(c_0) - \mu - \lambda = 0 \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial c_1}: \quad \beta u'(c_1) - \frac{\mu}{1+r} = 0. \tag{2}$$

The additional Kuhn-Tucker conditions are

$$\lambda[w_0 - c_0] = 0, \quad w_0 - c_0 \geq 0, \quad \text{and} \quad \lambda \geq 0. \tag{3}$$

The first condition in (3): complementary slackness condition.

Interior solution ($\lambda = 0$) yields the **Euler equation**:

$$u'(c_0) = \beta(1 + r)u'(c_1)$$

- LHS: marginal cost of saving one unit (less consumption today)
- RHS: discounted marginal benefit ($(1 + r)$ units tomorrow, valued at $\beta u'(c_1)$ each)
- When $\beta(1 + r) = 1$: full smoothing, $c_0 = c_1 = (w_0 + \beta w_1)/(1 + \beta)$

Generic T-Period Model: Result 1

If h is nonlinear, one cannot construct an intertemporal budget constraint (ibc), so we will use $x_{t+1} = h(x_t, y_t) \rightsquigarrow y_t = \widehat{h}(x_t, x_{t+1})$
Substitute the constraint into the objective: $F(x_t, x_{t+1}) \equiv \widehat{\mathcal{F}}(\widehat{h}(x_t, x_{t+1}))$.

$$\max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t F(x_t, x_{t+1}).$$

Result 1: If F is jointly concave in (x_t, x_{t+1}), increasing in x_t, decreasing in x_{t+1}, and the interior conditions hold, then the optimum satisfies:
Euler equation:

$$F_2(x_t^\star, x_{t+1}^\star) + \beta F_1(x_{t+1}^\star, x_{t+2}^\star) = 0 \qquad \forall t < T$$

Terminal condition: $x_{T+1}^\star = \underline{x}$

- $T + 2$ equations (Euler + initial + terminal), $T + 2$ unknowns
- Second-order difference equation in the state variable
- "Variational" condition: given x_t and x_{t+2}, vary x_{t+1} to achieve the best outcome

$$\max_{\{k_{t+1}\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t u(f(k_t) + (1-\delta)k_t - k_{t+1}) \tag{P3}$$

$$\text{subject to} \quad k_{t+1} \geq 0 \quad \forall t \leq T.$$

The Lagrangian associated to our maximization problem (P3) is

$$\mathcal{L} = \sum_{t=0}^{T} \beta^t \left\{ u(f(k_t) + (1-\delta)k_t - k_{t+1}) + \mu_t k_{t+1} \right\},$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial k_{t+1}}: \quad -u'(c_t) + \beta u'(c_{t+1}) \left[f'(k_{t+1}) + 1 - \delta \right] + \mu_t = 0, \quad t \in \{0, \ldots, T-1\}$$

$$\frac{\partial \mathcal{L}}{\partial k_{T+1}}: \quad -u'(c_T) + \mu_T = 0.$$

Kuhn-Tucker conditions also include $\mu_t k_{t+1} = 0$, $k_{t+1} \geq 0$, and $\mu_t \geq 0$, $0 \leq t \leq T$.

Finite-Horizon NGM: Euler Equation

FOC ($t = 0, \ldots, T-1$):

$$u'(c_t) = \beta u'(c_{t+1})[f'(k_{t+1}) + 1 - \delta] + \mu_t$$

Terminal period: $u'(c_T) = \mu_T > 0 \Rightarrow k_{T+1} = 0$.
Interior Euler equation (Inada $\Rightarrow k_{t+1} > 0$ for $t < T$, so $\mu_t = 0$):

$$\underbrace{u'(c_t)}_{\substack{\text{marginal cost} \\ \text{of investment}}} = \underbrace{\beta\, u'(c_{t+1})}_{\substack{\text{utility increase} \\ \text{per unit return}}} \cdot \underbrace{\left[f'(k_{t+1}) + 1 - \delta\right]}_{\text{return on investment}}.$$

Solving by Backward Induction (Example 4.1)

Log utility, Cobb-Douglas $f(k) = Ak^\alpha$, full depreciation $\delta = 1$.
Euler equation for $t < T$ becomes

$$\frac{1}{Ak_t^\alpha - k_{t+1}} = \beta \frac{\alpha A k_{t+1}^{\alpha-1}}{Ak_{t+1}^\alpha - k_{t+2}}.$$

Start from $k_{T+1} = 0$, use Euler equation backwards:

$$k_T = \frac{\alpha\beta}{1+\alpha\beta} Ak_{T-1}^\alpha, \qquad k_{T-1} = \frac{\alpha\beta(1+\alpha\beta)}{1+\alpha\beta+(\alpha\beta)^2} Ak_{T-2}^\alpha, \quad \dots$$

General solution:

$$k_{t+1} = \alpha\beta \cdot \frac{1-(\alpha\beta)^{T-t}}{1-(\alpha\beta)^{T-t+1}} \cdot Ak_t^\alpha$$

Saving rate: $s_t = \alpha\beta \cdot \frac{1-(\alpha\beta)^{T-t}}{1-(\alpha\beta)^{T-t+1}}$ – depends on time (distance to T), but not on k_t. ($\lim_{T\to\infty} s_t = \alpha\beta$, constant as in SOLOW)

CRRA Utility and the EIS

Constant Relative Risk Aversion (CRRA) utility:

$$u(c) = \frac{c^{1-\sigma} - 1}{1 - \sigma}, \qquad \sigma > 0,\ \sigma \neq 1.$$

Special cases: $\sigma = 0$ (linear), $\sigma \to 1$ (log), $\sigma \to \infty$ (Leontief).
Elasticity of Intertemporal Substitution (EIS): How does c_{t+s}/c_t respond to a change in the return $R_{t,t+s}$?
With CRRA and linear production $f_t(k) = R_t k$:

$$\frac{c_{t+s}}{c_t} = (\beta^s)^{1/\sigma} (R_{t,t+s})^{1/\sigma} \quad \Rightarrow \quad \text{EIS} = \frac{\partial \log(c_{t+s}/c_t)}{\partial \log R_{t,t+s}} = \frac{1}{\sigma}$$

- $\sigma < 1$: substitution effect dominates; investment rises with returns
- $\sigma > 1$: income effect dominates; investment falls with returns
- $\sigma = 1$ (log): income and substitution effects exactly cancel

4. Sequential Methods: Infinite Horizon

Moving to Infinite Horizon

Advantage: Problem becomes **stationary** – the maximization at t is the same as at $t+1$ for the same capital level.

Dynastic interpretation: β^t = weight on descendants t generations ahead. $\beta < 1$: care more about self than offspring.

Mathematical complications:

1. **Unbounded utility**: Need $\beta < 1$, and with growth at rate γ, need $\beta(1+\gamma)^{1-\sigma} < 1$
2. **Constraint sets "too large"**: Need borrowing limits to prevent infinite debt accumulation
   - × $a_{t+1} \geq \underline{a}$ may be too restrictive (why would that be for *every* period).
   - × A terminal condition $a_{T+1} \geq \underline{a}$ is an alternative.

A Ponzi-Game

- Without a terminal period, need to rule out schemes where agent borrows and never repays.
- Assume. consumer is given initial asset $a_0 > 0$ and has no income. Period budget constraint ($r > 0$

$$c_t + a_{t+1} = (1 + r)a_t$$

- Consider a candidate solution to the consumer's maximization problem $\{c_t^*\}_{t=0}^{\infty}$. Absent further constraints, the agent could improve on $\{c_t^*\}_{t=0}^{\infty}$ as follows:
  1. Let $\widetilde{c}_0 = c_0^* + \epsilon$, with $\epsilon > 0$, making $\widetilde{a}_1 = a_1^* - \epsilon$.
  2. For every $t \geq 1$ leave $\widetilde{c}_t = c_t^*$ by setting $\widetilde{a}_{t+1} = a_{t+1}^* - \epsilon(1 + r)^t$.
- Strictly increasing utility function $\rightsquigarrow$ the agent is clearly better off and satisfies the budget constraint period by period.
- This sort of improvement is possible for any candidate solution $\rightsquigarrow$ no maximum for lifetime utility.
- And the agent's debt is growing without bound at rate $1 + r$, and it is never repaid.
- This type of scheme, borrowing $(1 + r)a_t$ every period t to keep rolling the debt, is often called a "Ponzi scheme" or a "Ponzi game".

The No-Ponzi-Game Condition

- **No-Ponzi-game (nPg) condition**:

$$\lim_{t\to\infty} \frac{a_{t+1}}{(1+r)^t} \geq 0$$

  Agent cannot have positive debt (in present value) at infinity.
- Take $c_0 + a_1 = (1+r)a_0$ and $c_1 + a_2 = (1+r)a_1$. This implies $c_0 + \frac{c_1}{1+r} = a_0 + \frac{a_2}{1+r}$.
- Substitute a_2 forward, etc...

$$\sum_{t=0}^{T} \frac{c_t}{(1+r)^t} = a_0(1+r) - \frac{a_{T+1}}{(1+r)^T}.$$

- Taking limits, and using the nPg, we arrive at

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = a_0(1+r) - \lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^T} \leq a_0(1+r),$$

Lifetime budget constraint

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} \leq a_0(1+r)$$

- Ruling out Ponzi schemes is a natural infinite-horizon extension of the constraint $a_{T+1} \geq 0$ in a finite-horizon model.
- The "natural borrowing limit":
  - × Both in the finite-horizon and the infinite-horizon cases
  - × The loosest possible constraint on borrowing
  - × The only restriction is that you are able to pay back, in a present-value sense, if you set consumption to zero at all future times.
  - × In the particular model here, the natural borrowing limit is zero at all times
- The nPg condition and the lifetime budget constraint (and the natural borrowing limit, in this case $a_t \geq 0$ for all t) are equivalent.

The Transversality Condition

- **The transversality condition (TVC)**:

$$\lim_{t\to\infty} \frac{a_{t+1}}{(1+r)^t} \leq 0$$

- **Interpretation**: It is never optimal to hold positive assets (in present value) at infinity – better to consume them.
- **nPg vs. TVC**:
  - × **nPg**: Externally imposed. Prevents over-*borrowing*.
  - × **TVC**: Self-imposed optimality condition.
- In the NGM, the TVC takes the form:

$$\lim_{t\to\infty} \beta^t u'(c_t)[f'(k_t) + 1 - \delta]k_t = 0$$

- In the NGM, if the economy converges to a steady state with $\beta < 1$: TVC satisfied automatically.

Sufficiency: Proposition 4.4

Proposition: If $F(x_t, x_{t+1})$ is jointly concave, Γ is convex, and $x_t^\star$ satisfies:
(i) Euler equation: $F_2(x_t^\star, x_{t+1}^\star) + \beta F_1(x_{t+1}^\star, x_{t+2}^\star) = 0$ for all t
(ii) TVC: $\lim_{t\to\infty} \beta^t F_1(x_t^\star, x_{t+1}^\star) x_t^\star = 0$
then $\{x_t^\star\}$ maximizes the objective.

Proof strategy (Appendix 4.A.3): Use the Euler equation repeatedly plus the definition of concavity ($f(x) \leq f(x^\star) + f'(x^\star)(x - x^\star)$) to show that any alternative feasible sequence gives a lower objective value – provided the TVC holds.

Solving the Infinite-Horizon NGM (Example 4.5)

Log utility, $f(k) = Ak^\alpha$, $\delta = 1$. Take the limit of the finite-horizon solution as $T \to \infty$:

$$k_{t+1} = \lim_{T\to\infty} \alpha\beta \cdot \frac{1-(\alpha\beta)^{T-t}}{1-(\alpha\beta)^{T-t+1}} \cdot Ak_t^\alpha = \alpha\beta A k_t^\alpha$$

- Saving rate $s = \alpha\beta$ (constant – no longer depends on time!)
- Stationarity: infinite horizon $\Rightarrow$ same problem every period
- TVC: $\lim_{t\to\infty} \beta^t \alpha/(1-\alpha\beta) = 0$ since $\beta < 1$ ✓

Steady state: $\overline{k} = (\beta\alpha A)^{1/(1-\alpha)}$ – identical to SOLOW with $s = \alpha\beta$ and $\delta = 1$.

Guess and verify (Example 4.6): Guess $k_{t+1} = sAk_t^\alpha$, substitute into Euler equation $\Rightarrow s = \alpha\beta$.

Steady State (no growth)

- From the Euler equation with $c_t = c_{t+1}$:

$$1 = \beta(f'(\overline{k}) + 1 - \delta)$$

- Unique solution under Inada conditions. This is the **modified golden rule**: $f'(\overline{k}) = 1/\beta - 1 + \delta$.

Balanced Growth

- $Y_t = F(K_t, A_t L_t)$, A and L grow at rates γ and n
- CRRA preferences $u(c_t) = c_t^{1-\sigma}/(1-\sigma)$, $\sigma > 0$ and $\sigma \neq 1$.
- Utilitarian Social Planner: with $c = C/L$, objective is

$$\sum_{t=0}^{\infty} \beta^t L_t u(c_t),$$

- The resource constraint with growth is

$$C_t = F(K_t, A_t L_t) + (1-\delta)K_t - K_{t+1}.$$

- With "per efficiency units of labor" variables: $\widetilde{x}_t = X_t/(A_t L_t)$,

$$\widetilde{c}_t = f(\widetilde{k}_t) + (1-\delta)\widetilde{k}_t - (1+\gamma)(1+n)\widetilde{k}_{t+1}.$$

- Social Planner cares about consumption per capita:
  define $\widetilde{k}_t = K_t/(A_t L_t)$.
  Adjusted discount factor: $\widetilde{\beta} \equiv \beta(1+n)(1+\gamma)^{1-\sigma}$

Balanced Growth

- Normalizing $A_0 = 1$ and $L_0 = 1$, we can rewrite the objective function as

$$\sum_{t=0}^{\infty} \beta^t L_t \frac{c_t^{1-\sigma}}{1-\sigma} = \sum_{t=0}^{\infty} \beta^t (1+n)^t \frac{\left((1+\gamma)^t \widetilde{c}_t\right)^{1-\sigma}}{1-\sigma}$$
$$= \sum_{t=0}^{\infty} \widetilde{\beta}^t \frac{\widetilde{c}_t^{1-\sigma}}{1-\sigma},$$

  where now we use the adjusted discount factor $\widetilde{\beta}$:

$$\widetilde{\beta} \equiv \beta(1+n)(1+\gamma)^{1-\sigma}.$$

- Euler equation on BGP:

$$(1+n)(1+\gamma) = \widetilde{\beta}[f'(\overline{\widetilde{k}}) + 1 - \delta]$$

- **CRRA is the only utility consistent with balanced growth**.

5. Recursive Methods: Dynamic Programming

- **Key idea**: Break the infinite-dimensional problem into a sequence of identical small problems.
- Take advantage of stationnarity
- Take the infinite-horizon version of the problem (P1):

$$V(x_0) \equiv \max_{\{x_{t+1} \in \Gamma(x_t)\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t F(x_t, x_{t+1}),$$

and break it into pieces:

The Bellman Equation

$$
\begin{aligned}
V(x_0) &= \max_{\{x_{t+1}\in\Gamma(x_t)\}_{t=0}^{\infty}} \left\{ F(x_0, x_1) + \beta \sum_{t=1}^{\infty} \beta^{t-1} F(x_t, x_{t+1}) \right\} \\
&= \max_{x_1\in\Gamma(x_0)} \left\{ F(x_0, x_1) + \beta \left[\max_{\{x_{t+1}\in\Gamma(x_t)\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} F(x_t, x_{t+1}) \right] \right\} \\
&= \max_{x_1\in\Gamma(x_0)} \left\{ F(x_0, x_1) + \beta \left[\underbrace{\max_{\{x_{t+2}\in\Gamma(x_{t+1})\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^{t} F(x_{t+1}, x_{t+2})}_{\equiv V(x_1)} \right] \right\}
\end{aligned}
$$

The Bellman Equation

$$V(x) = \max_{x' \in \Gamma(x)} \{F(x, x') + \beta V(x')\}$$

This is the **Bellman equation** – a functional equation (unknown is V, a function).
Policy function: $g(x) = \arg\max_{x' \in \Gamma(x)} \{F(x, x') + \beta V(x')\}$
$\Rightarrow x' = g(x)$: stationary decision rule.

The NGM in Recursive Form

The standard NGM:

$$V(k) = \max_{k' \in [0, f(k)+(1-\delta)k]} \{u(f(k) + (1-\delta)k - k') + \beta V(k')\}$$

State: k (current capital). **Control**: k' (or equivalently c).

Other examples:

- **Periodic NGM** (alternating TFP A_h, A_l): Two value functions $V_h(k), V_l(k)$.
- **Delayed depreciation NGM** (capital depreciates after two periods): Two state variables (i_{-1}, i_{-2}).

Key lesson: Choosing state variables appropriately is essential for writing the problem recursively.

Properties of the Value Function

Under standard assumptions (F continuous, concave; Γ nonempty, compact, convex; $\beta < 1$):

1. **Contraction mapping**: Starting from any V_0, iterating $V_{n+1}(x) = \max_{x'}\{F(x, x') + \beta V_n(x')\}$ converges to V. Rate: $\|V_{n+1} - V\| \leq \beta \|V_n - V\|$.
2. V is **strictly increasing** (if F increasing in first argument and Γ monotone)
3. V is **strictly concave** (if F jointly strictly concave and Γ convex) $\Rightarrow$ policy g is unique
4. V is **differentiable** at interior solutions (if F is C^1)
5. g is **strictly increasing** (under all above conditions)

Property 1 is crucial for numerical computation (value function iteration, Chapter 10).

Solving the Bellman Equation (Example 4.7)

- Log utility, $f(k) = Ak^\alpha$, $\delta = 1$.
- Bellman equatio is

$$V(k) = \max_{k' \geq 0} \{\log(Ak^\alpha - k') + \beta V(k')\}.$$

- Start from $V_0(k) = 0$:

$$V_1(k) = \max_{k'} \log(Ak^\alpha - k') + V_0(k') = \log A + \alpha \log k \quad \text{(optimal: } k' = 0)$$

$$V_2(k) = \max_{k'} \{\log(Ak^\alpha - k') + \beta[\log A + \alpha \log k']\}$$

- FOC:

$$\frac{1}{Ak^\alpha - k'} = \frac{\beta\alpha}{k'} \implies k' = \frac{\alpha\beta Ak^\alpha}{1 + \alpha\beta}.$$

Solving the Bellman Equation (Example 4.7)

- Plug into the definition of V_2:

$$V_2(k) = \log\left(Ak^\alpha - \frac{\alpha\beta Ak^\alpha}{1+\alpha\beta}\right) + \beta\left[\log A + \alpha \log \frac{\alpha\beta Ak^\alpha}{1+\alpha\beta}\right]$$

$$= \underbrace{\log\left(A - \frac{\alpha\beta A}{1+\alpha\beta}\right) + \beta \log A + \alpha\beta \log \frac{\alpha\beta A}{1+\alpha\beta}}_{=a_2} + \underbrace{\left(\alpha + \alpha^2\beta\right)}_{=b_2} \log k.$$

- Repeat and V_n will converge towards

$$V = \frac{1}{1-\beta} \cdot \frac{1}{1-\alpha\beta}\left[\log A + \log(1-\alpha\beta)^{1-\alpha\beta} + \log(\alpha\beta)^{\alpha\beta}\right] + \frac{\alpha}{1-\alpha\beta} \log k$$

Solving the Bellman Equation (Example 4.7)

- Alternative way: **Guess and verify**
- Pattern: $V_n(k) = a_n + b_n \log k$ for all n.
- Guess $V(k) = a + b \log k$:
- Bellman:

$$V(k) = \max_{k' \geq 0} \{ \log(Ak^\alpha - k') + \beta(a + b \log k') \}.$$

- FOC:

$$\frac{1}{Ak^\alpha - k'} = \frac{\beta b}{k'}$$

$\rightsquigarrow k' = \alpha\beta A k^\alpha$.

- Therefore

$$V(k) = \log(Ak^\alpha - \alpha\beta A k^\alpha) + \beta(a + b \log(\alpha\beta A k^\alpha))$$

- Solution:

$$b = \frac{\alpha}{1 - \alpha\beta}, \qquad a = \frac{1}{1-\beta} \frac{1}{1-\alpha\beta} \left[\log A + \log(1 - \alpha\beta)^{1-\alpha\beta} (\alpha\beta)^{\alpha\beta} \right]$$

- Policy: $k' = \alpha\beta A k^\alpha$ – same answer as sequential methods.

The Functional Euler Equation and the Envelope Theorem

- Bellman equation:
$$V(x) = \max_{x' \in \Gamma(x)} \{F(x, x') + \beta V(x')\}.$$
- Policy function $x' = g(x)$. Put in Bellman equation:
$$V(x) = F(x, g(x)) + \beta V(g(x)).$$
- Assuming an interior solution, x' satisfies the first-order condition
$$F_2(x, x') + \beta V'(x') = 0.$$
- Evaluated at the optimum, i.e., at $x' = g(x)$,
$$F_2(x, g(x)) + \beta V'(g(x)) = 0.$$
- In contrast to how we derived the sequential formulation, we now need to know the derivative of the value function $V'(\cdot)$ in order to solve for the policy rule.

The Functional Euler Equation and the Envelope Theorem

- Differentiate
$$V(x) = F(x, g(x)) + \beta V(g(x)).$$
- gives
$$V'(x) = F_1(x, g(x)) + g'(x) \underbrace{\left[F_2(x, g(x)) + \beta V'(g(x))\right]}_{\text{indirect effect through optimal choice of } x'},$$
- We have shown before that at the optimum
$$F_2(x, g(x)) + \beta V'(g(x)) = 0$$
- The indirect effect disappears (Envelope Theorem)
- We then have
$$V'(x) = F_1(x, g(x))$$

The Functional Euler Equation and the Envelope Theorem

$$\begin{aligned} F_2(x, g(x)) + \beta V'(g(x)) &= 0 && \text{(First-order condition)} \\ V'(x) &= F_1(x, g(x)) && \text{(Envelope condition)} \end{aligned}$$

- $V'(g(x)) = F_1(g(x), g(g(x)))$
- **Functional Euler equation**:

$$F_2(x, g(x)) + \beta F_1(g(x), g(g(x))) = 0 \qquad \forall x$$

or

$$F_2(x_t, x_{t+1}) + \beta F_1(x_{t+1}, x_{t+2}) = 0, \quad \forall t,$$

- × Same as the sequential Euler equation, but written as a **functional equation** in g
- × Unknown is the function g (not a sequence)
- × Necessary condition only (may have multiple solutions); the Bellman equation selects the correct one
- × Useful for numerical computation and for proving convergence

6. Dynamics in the NGM

Maximum Attainable Level of k

- There is a unique maximum attainable level of the capital stock, k_u.
- If the capital stock starts out below (or at) k_u, it cannot take a higher value than k_u in the future.
- Why? For any k, $f(k) + (1 - \delta)k$ is the highest feasible saving k' (0 consumption).
- k_u defined by $f(k_u) + (1 - \delta)k_u = k_u$ (discard solution $k = 0$
- We can restrict attention to a search for value and policy functions over the closed and bounded set $[0, k_u]$.

- The standard NGM:

$$V(k) = \max_{k' \in [0, k_u]} \{u(f(k) + (1-\delta)k - k') + \beta V(k')\}$$

- With $g(k) = k'$, Euler equation is:

$$F_2(x, g(x)) + \beta F_1(g(x), g(g(x))) = 0 \qquad \forall x$$

which is here

$$u'\big(f(k)+(1-\delta)k-g(k)\big) = \beta\, u'\big(f(g(k))+(1-\delta)g(k)-g(g(k))\big)\big(f'(g(k))+1-\delta\big).$$

- Note that steady state solves

$$1 = \beta\big(f'(\bar{k}) + 1 - \delta\big)$$

Proof of Global Convergence: Envelope Condition

- **Goal**: Show that $k_t \to \overline{k}$ from any $k_0 > 0$.
- Under the standard assumptions, $V(k)$ is strictly concave:

$$\big[V'(k) - V'(g(k))\big]\big[k - g(k)\big] \leq 0 \quad \forall k \in [0, k_u].$$

- Envelope Condition:

$$V'(k) = u'\big(f(k) + (1-\delta)k - g(k)\big)\big(f'(k) + 1 - \delta\big).$$

Proof of Global Convergence: the First-Order Condition

- Bellman equation of the growth model, with $c = f(k) + (1-\delta)k - k'$:

$$V(k) = \max_{k' \in [0, k_u]} \Big\{ u\big(f(k) + (1-\delta)k - k'\big) + \beta V(k') \Big\},$$

  i.e. $F(k, k') = u\big(f(k) + (1-\delta)k - k'\big)$, so $F_2(k, k') = -u'(c)$.
- First-order condition $F_2(k, k') + \beta V'(k') = 0$:

$$-u'(c) + \beta V'(k') = 0.$$

- Evaluated at the optimum $k' = g(k)$ and solved for V':

$$V'(g(k)) = \frac{1}{\beta}\, u'\big(f(k) + (1-\delta)k - g(k)\big).$$

- Both $V'(k)$ and $V'(g(k))$ are now proportional to the same $u'(c) \Rightarrow$ we can factorize $u'(c)$ in the concavity inequality.

Proof of Global Convergence

- Inserting these two expressions into the left-most bracket of the inequality and factorizing u', we obtain

$$u'\big(f(k)+(1-\delta)k-g(k)\big)\left[f'(k)+1-\delta-\frac{1}{\beta}\right]\big[k-g(k)\big]\leq 0 \quad \forall k\in[0,k_u]. \quad (4)$$

  - × $k=\overline{k}$: $f'(\overline{k})+1-\delta=1/\beta$ and $g(\overline{k})=\overline{k}$ ✓
  - × $k>\overline{k}$: $f'(k)+1-\delta<1/\beta \Rightarrow g(k)<k$ (capital falls)
  - × $k<\overline{k}$: $f'(k)+1-\delta>1/\beta \Rightarrow g(k)>k$ (capital rises)
- We have global monotonic convergence

Local Approximation of the Policy Function

- Approximate $g(k) \approx a_0 + a_1 k$ near $\overline{k}$.
- From $g(\overline{k}) = \overline{k}$: $a_0 = \overline{k}(1 - a_1)$ $\rightsquigarrow$ one restriction for a_0 and a_1
- Functional Euler equation:

$$u'\big(f(k)+(1-\delta)k-g(k)\big) = \beta\, u'\big(f(g(k))+(1-\delta)g(k)-g(g(k))\big)\big(f'(g(k))+1-\delta\big).$$

- Differentiate the functional Euler equation w.r.t. k, evaluate at $\overline{k}$:

$$\mathrm{LHS}_k(\bar{k}) = u''(\bar{c})\left[\underbrace{f'(\bar{k}) + 1 - \delta}_{1/\beta} - \underbrace{g'(\bar{k})}_{=a_1}\right]$$

$$\mathrm{RHS}_k(\bar{k}) = \beta u''(\bar{c})\left[(f'(\bar{k}) + 1 - \delta)a_1 - \underbrace{g'(\bar{k})g'(\bar{k})}_{=a_1^2}\right]\left[f'(\bar{k}) + 1 - \delta\right] + \beta u'(\bar{c}) f''(\bar{k}) a_1.$$

- We then have

$$u''(\overline{c})[1 - \beta a_1] = u''(\overline{c})[a_1 - \beta a_1^2] + \beta^2 u'(\overline{c}) f''(\overline{k}) a_1$$

Local Approximation of the Policy Function

$$u''(\overline{c})[1-\beta a_1] = u''(\overline{c})[a_1 - \beta a_1^2] + \beta^2 u'(\overline{c}) f''(\overline{k}) a_1$$

- This is a quadratic equation in a_1
- Define $\Theta \equiv \beta^2 u'(\overline{c}) f''(\overline{k}) / u''(\overline{c}) > 0$. Solving the quadratic:

$$a_1 = \frac{1+\beta+\Theta - \sqrt{(1+\beta+\Theta)^2 - 4\beta}}{2\beta}$$

- With CRRA, $u(c) = (c^{1-\sigma}-1)/(1-\sigma)$:

$$\Theta = -\frac{\beta^2 f''(\bar{k})\big(f(\bar{k}) - \delta\bar{k}\big)}{\sigma}.$$

  - × $a_1 \in (0,1]$: the correct root (monotone convergence)
  - × a_1 decreasing in Θ (faster convergence when Θ is large)
  - × σ large (unwilling to change c) $\Rightarrow \Theta$ small $\Rightarrow a_1$ close to 1 $\Rightarrow$ slow convergence
  - × $\sigma = 0$ (linear utility) $\Rightarrow \Theta \to \infty \Rightarrow a_1 \to 0 \Rightarrow$ immediate convergence

Calibration: The Role of σ

With CRRA $u(c) = (c^{1-\sigma} - 1)/(1-\sigma)$ and Cobb-Douglas $f(k) = k^{\alpha}$:

$$\Theta = \frac{\beta^2(1-\alpha)}{\sigma}\left(\frac{1}{\beta} - 1 + \delta\right)\left(\frac{1}{\alpha}\left(\frac{1}{\beta} - 1 + \delta\right) - \delta\right)$$

Θ is positive, decreasing in σ, decreasing in α, increasing in δ.

Empirical estimates of $1/\sigma$ (EIS):

- Hall (1988), aggregate data: $1/\sigma \approx 0.1$
- Attanasio and Weber (1993, 1995), micro data: $1/\sigma \in [0.3, 0.8]$
- Most applied macro: $\sigma = 1$ (log) or $\sigma = 2$

Result: Convergence in the NGM is qualitatively like Solow but quantitatively different – the endogenous saving rate depends on k through σ.

7. Summary

Key Takeaways (I)

1. **Generic problem**: Controls y_t, states x_t, discount factor β, feasibility Γ
2. **Euler equation**: $F_2(x_t, x_{t+1}) + \beta F_1(x_{t+1}, x_{t+2}) = 0$ – marginal cost of saving = marginal benefit
3. **Terminal condition** (finite horizon): $x_{T+1} = \underline{x}$ (consume everything)
4. **nPg condition** (infinite horizon): $\lim a_{t+1}/(1+r)^t \geq 0$ (no exploding debt)
5. **TVC**: $\lim \beta^t F_1 x_t = 0$ (no over-accumulation); self-imposed optimality condition

Key Takeaways (II)

6. **CRRA utility**: EIS $= 1/\sigma$; the only utility consistent with balanced growth
7. **Bellman equation**: $V(x) = \max_{x'}\{F(x, x') + \beta V(x')\}$; contraction mapping $\Rightarrow$ unique V, can be found by iteration
8. **Properties of** V: increasing, strictly concave, differentiable at interior; policy g unique and increasing
9. **Functional Euler equation**: $F_2(x, g(x)) + \beta F_1(g(x), g(g(x))) = 0$; necessary condition from Envelope Theorem
10. **Global convergence**: Proved using strict concavity of V + Envelope Theorem; speed depends on σ (curvature of utility) and α (curvature of production)
11. **Local approximation**: $g(k) \approx a_0 + a_1 k$; a_1 solved from a quadratic; $a_1 \in (0, 1)$ for the stable root

